BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 114 · String Manipulation

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. Which CPU component performs arithmetic and logical operations?

Q2. Why might an HDD suit a large archive?
A. No moving parts
B. Low cost per GB
C. Unlimited lifespan
D. Always fastest access

Q3. “An SSD is faster because its magnetic read head rotates more quickly.” Correct this. Explain why.

Q4. A laptop is used on a bumpy bus. “Portability and durability mean the same thing.” Correct this. How should this affect the storage choice?

Do Now Recall previous learning | **I Do** Read new knowledge | **We Do** Practise with guidance | **You Do** Work independently | **Review & Improve** Check understanding

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

Concatenation — Joining strings creates new text; it does not add numerical values.

Length — Counting characters helps validate usernames and input.

Slicing — A slice extracts a specified portion of a string, using the selected language's indexing rules.

Data type — Characters and strings represent text, even when they contain numeric-looking digits.

New learning

Concatenation: Joining strings creates new text; it does not add numerical values.

Length: Counting characters helps validate usernames and input.

Slicing: A slice extracts a specified portion of a string, using the selected language's indexing rules.

Data type: Characters and strings represent text, even when they contain numeric-looking digits.

Worked example

A school prints a username from firstName = 'Sam' and lastName = 'Lee'. Explain how to produce 'Sam.Lee'.

1. Identify both strings and the separator.

2. Concatenate firstName + '.' + lastName.

3. The result is the string 'Sam.Lee'; do not add the text mathematically.

4. Check that the period occurs once and no spaces were introduced.

We Do · Practise with guidance

Guided 1

A library creates an ID from 'BK' and '204'. What string is produced?

Worked method: 'BK204'.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

2. Write a complete new answer in GREEN PEN. Show working if needed:

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Code First · Predict → Write → Test → Improve

Complete the OCR string operation to display NETWORK.

```
word = "network"
print(word....)
```

write

Question 1

State the difference between concatenation and arithmetic addition. [2 marks]

Question 2

A user enters 'AB' and '12'. Write the resulting concatenated string with no spaces. [1 mark]

Question 3

Explain why checking string length can support validation of a password. [3 marks]

Exit Ticket · Answer before checking

Concatenating strings always adds their numeric values.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. The ALU performs arithmetic and logical operations.

Q2. B — Low cost per GB

Q3. It does not need a moving head to seek positions on spinning platters.

Q4. Durability means resisting damage; portability means being easy to carry. An SSD has no moving read/write parts, so it handles shocks better. Size and weight also matter.

I Do / We Do pages · Guided practice

Q1. 'BK204'.

You Do pages · Code First

```
word = "network"
print(word.upper)
```

Expected result: NETWORK

You Do pages · Independent questions

Q1. Concatenation joins string content; arithmetic addition calculates a numerical sum.

Q2. AB12.

Q3. Length checks whether the input meets a stated minimum/maximum number of characters; alone it does not guarantee a secure password.

Exit Ticket · Model answer

False

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question: